

# Core ML: Modular Long-Context Sequence Modeling

Ansh

SAiDL Summer Assignment 2026

## 1 Overview

This report studies long-context sequence modeling on WikiText-2 using a modular Transformer language model. The codebase allows independent swaps of attention mechanisms, positional encodings, and convolution-attention hybrid blocks.

| Assignment Component | Coverage                                                                                                                                                                                                                          |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Baseline Transformer | Standard causal Transformer trained at context length 1024.                                                                                                                                                                       |
| Attention variants   | Standard, sliding-window/local, linear kernelized, MQA, and GQA are implemented; standard, sliding-window, linear, and GQA are compared at 1024 tokens for perplexity and at 512/1024/2048 tokens for latency and memory scaling. |
| Positional encodings | Sinusoidal, RoPE, ALiBi, and learned relative position bias are implemented and evaluated.                                                                                                                                        |
| Extrapolation        | Sinusoidal, RoPE, ALiBi, and relative-bias models trained at 512 tokens and evaluated at 512, 1024, and 2048 tokens.                                                                                                              |
| Hybrid blocks        | Conv-before-attention, interleaved convolution, and gated-convolution FFN are implemented; conv-before and gated-conv are evaluated with the best GQA + ALiBi configuration.                                                      |
| Logging              | JSONL/W&B logging, validation perplexity, throughput, peak memory, epoch timing, gradient norm, and final metric artifacts are included.                                                                                          |

Table 1: Core ML requirement coverage.

## 2 Baseline

### 2.1 Architecture

The baseline is a pre-normalized causal Transformer language model with tied token embedding and language-model head weights. The default configuration uses a context length of 1024 tokens, embedding dimension 512, 8 heads, 6 layers, and a feed-forward dimension of 2048.

### 2.2 Training Setup

- Dataset: WikiText-2 raw.
- Tokenizer: GPT-2 tokenizer.
- Optimizer: AdamW.
- Model width/depth: 512 hidden dimension, 8 attention heads, 6 Transformer layers, and feed-forward dimension 2048.

- Sequence length: 1024 tokens for the initial baseline and attention-variant experiments.
- Batch size: 4.
- Logged metrics: training loss, validation loss, validation perplexity, peak GPU memory, throughput, and elapsed time.

| Model                 | Context | Val Loss | Val PPL | Val Tokens/sec | Train Peak Mem (MB) |
|-----------------------|---------|----------|---------|----------------|---------------------|
| Standard + Sinusoidal | 1024    | 5.862    | 351.41  | 29636.66       | 4182.75             |

Table 2: Baseline results.

The standard Transformer with sinusoidal positional encoding serves as the reference point for all later comparisons. At step 8500 it reaches a validation loss of 5.862 and validation perplexity of 351.41 while maintaining high throughput and moderate memory use. This baseline is strong enough to act as a meaningful control but still leaves ample room for architectural improvement.

### 3 Attention Variants

The implemented attention variants are standard causal self-attention, sliding window attention, linear kernelized attention, multi-query attention, and grouped-query attention.

| Attention      | Context | Step | Val Loss | Val PPL | Val Tokens/sec | Train Peak Mem (MB) |
|----------------|---------|------|----------|---------|----------------|---------------------|
| Standard       | 1024    | 8500 | 5.862    | 351.41  | 29636.66       | 4182.75             |
| Sliding Window | 1024    | 5500 | 6.052    | 424.96  | 28301.10       | 4188.65             |
| Linear         | 1024    | 8500 | 5.856    | 349.49  | 23573.86       | 4981.47             |
| GQA            | 1024    | 8500 | 5.810    | 333.78  | 29830.34       | 4155.67             |

Table 3: Attention ablation results.

Among the attention variants evaluated so far, grouped-query attention (GQA) performs best. It achieves the lowest validation loss and validation perplexity while also remaining efficient in both throughput and memory. Linear attention slightly improves over the standard baseline in validation perplexity, but this comes with noticeably higher memory consumption and lower throughput. The sliding-window model performs worst in the current results, although that run was logged at a smaller training budget than the others and should therefore be treated as a preliminary comparison rather than a final conclusion.

The assignment suggests evaluating attention variants across multiple context lengths such as 512, 1024, 2048, and 4096 where feasible. Under the available compute budget, I prioritized full training comparisons at the main 1024-token context, then measured forward-pass latency and memory scaling for the main attention variants at 512, 1024, and 2048 tokens. A full retraining grid over every attention variant and every context length is therefore left as a compute-bound extension rather than a missing architectural component.

### 4 Positional Encodings

The implemented positional choices are sinusoidal positional encodings, rotary positional embeddings, ALiBi, and learned relative position bias. In addition to the main 1024-token context experiments, I evaluate length extrapolation by training the positional variants at  $L_{\text{train}} = 512$  and evaluating them at  $L_{\text{test}} \in \{512, 1024, 2048\}$ .

ALiBi is the strongest positional encoding tested so far in the current setup. The completed GQA + ALiBi run achieved its best logged validation performance at step 7000, reaching validation loss 4.985 and validation perplexity 146.24. The final checkpoint at step 8500 remained close, with validation loss 4.996 and validation perplexity 147.87, indicating only mild late-stage degradation.

| Position Method | Attention | Context | Step | Val PPL | Notes                                          |
|-----------------|-----------|---------|------|---------|------------------------------------------------|
| Sinusoidal      | GQA       | 1024    | 8500 | 333.78  | Fully completed run                            |
| RoPE            | GQA       | 1024    | 6500 | 150.28  | Best logged validation checkpoint              |
| RoPE (final)    | GQA       | 1024    | 8500 | 152.73  | Completed rerun, slight late-stage degradation |
| ALiBi           | GQA       | 1024    | 7000 | 146.24  | Best logged validation checkpoint              |
| ALiBi (final)   | GQA       | 1024    | 8500 | 147.87  | Completed run, slight late-stage degradation   |
| Relative Bias   | GQA       | 1024    | 8500 | 235.86  | Completed run, weaker than ALiBi/RoPE          |

Table 4: Current positional encoding results.

RoPE also substantially improves over sinusoidal positional encoding. The completed GQA + RoPE rerun achieved its best logged validation performance at step 6500, reaching validation loss 5.012 and validation perplexity 150.28. Although ALiBi slightly outperforms RoPE in the current experiments, both methods dramatically improve over the GQA + sinusoidal baseline. Relative positional bias also improves over sinusoidal, reaching validation perplexity 235.86, but it underperforms both ALiBi and RoPE by a wide margin. This suggests that the simple learned relative-bias formulation used here is less effective than the more structured extrapolation-friendly encodings in this setting.

## 4.1 Length Extrapolation

To directly test whether the positional encodings generalize beyond their training context, I trained models with sequence length 512 and evaluated their final checkpoints at longer validation contexts. The standard Transformer with sinusoidal encoding is included as the absolute-position baseline, and the GQA runs compare RoPE, ALiBi, and learned relative position bias.

| Method     | Eval Context | Val Loss | Val PPL | Eval Time (s) |
|------------|--------------|----------|---------|---------------|
| Sinusoidal | 512          | 5.7039   | 300.04  | 7.95          |
| Sinusoidal | 1024         | 6.2477   | 516.83  | 9.10          |
| Sinusoidal | 2048         | 7.1561   | 1281.94 | 11.16         |
| ALiBi      | 512          | 5.0253   | 152.21  | 8.17          |
| ALiBi      | 1024         | 5.0033   | 148.90  | 9.91          |
| ALiBi      | 2048         | 4.9956   | 147.76  | 13.29         |
| RoPE       | 512          | 5.0701   | 159.20  | 8.13          |
| RoPE       | 1024         | 5.2424   | 189.13  | 9.39          |
| RoPE       | 2048         | 5.6356   | 280.24  | 11.86         |
| Relative   | 512          | 5.2876   | 197.86  | 9.29          |
| Relative   | 1024         | 5.3719   | 215.27  | 12.04         |
| Relative   | 2048         | 5.5349   | 253.37  | 17.90         |

Table 5: Length extrapolation for models trained at context length 512.

The extrapolation result strongly favors ALiBi. Its validation perplexity remains stable when the evaluation context is extended beyond the training length, slightly improving from 152.21 at 512 tokens to 147.76 at 2048 tokens. RoPE is competitive at the training context, reaching 159.20 perplexity at 512 tokens, but degrades to 189.13 at 1024 tokens and 280.24 at 2048 tokens. Learned relative bias sits between ALiBi and RoPE at the longest context, reaching 253.37 perplexity at 2048 tokens, but is worse than both at the training length. All three long-context encodings are substantially better than the standard sinusoidal baseline, which collapses from 300.04 perplexity at 512 tokens to 1281.94 at 2048 tokens. This is consistent with ALiBi’s design: the positional signal is computed as a distance-dependent attention bias rather than relying on absolute position values, learned distance tables, or rotary phases that may not extrapolate as smoothly without additional scaling. The result supports using GQA + ALiBi as the final Core ML architecture, especially when length extrapolation is part of the evaluation criterion.

## 4.2 Latency Benchmark

I also benchmarked the main attention variants across sequence lengths to quantify the cost of longer-context evaluation. These measurements use dense forward passes with batch size 2 on the same GPU environment and are intended to compare architectural scaling rather than cached autoregressive decoding.

| Model          | Seq Len | Tokens/sec | Avg Latency (ms) | Peak Mem (MB) |
|----------------|---------|------------|------------------|---------------|
| Standard       | 512     | 33068.5    | 30.97            | 382.0         |
| Standard       | 1024    | 30230.8    | 67.75            | 581.1         |
| Standard       | 2048    | 23811.6    | 172.02           | 981.5         |
| Sliding Window | 512     | 34221.8    | 29.92            | 382.0         |
| Sliding Window | 1024    | 28788.1    | 71.14            | 581.1         |
| Sliding Window | 2048    | 22128.3    | 185.10           | 1007.5        |
| Linear         | 512     | 22563.5    | 45.38            | 577.7         |
| Linear         | 1024    | 22390.7    | 91.47            | 972.5         |
| Linear         | 2048    | 22307.9    | 183.61           | 1763.5        |
| GQA + ALiBi    | 512     | 31897.5    | 32.10            | 369.0         |
| GQA + ALiBi    | 1024    | 27673.8    | 74.01            | 568.1         |
| GQA + ALiBi    | 2048    | 20359.0    | 201.19           | 968.5         |

Table 6: Inference latency benchmark across attention variants with batch size 2. Standard, sliding-window, and linear rows use sinusoidal positions; the final selected model uses GQA + ALiBi.

The benchmark shows the expected growth in latency and memory with sequence length. Standard attention is fastest in this forward-only benchmark at 2048 tokens, while GQA + ALiBi uses fewer parameters and slightly less memory. Linear attention has almost flat tokens/sec across sequence lengths, but its current implementation is memory-heavy because the cumulative kernel statistics are materialized with an additional head-dimension product. Sliding-window attention is competitive at 512 tokens but does not improve the 2048-token forward-pass latency in this implementation. Taken together with the validation perplexity results, the selected GQA + ALiBi model remains the best quality-oriented final choice, even though standard attention remains a strong raw-throughput baseline.

## 5 Convolution-Attention Hybrids

Two hybrid designs are implemented: a depthwise convolution branch before each attention layer, and a gated convolutional feed-forward block. These are tested with the best attention and positional encoding from the earlier ablations.

| Block Type            | Val PPL | Tokens/sec | Peak Memory MB | Notes                               |
|-----------------------|---------|------------|----------------|-------------------------------------|
| Standard              | 146.24  | 28440.73   | 4153.57        | Best GQA + ALiBi checkpoint         |
| Conv Before Attention | 173.83  | 20122.53   | 6453.46        | Worse quality and efficiency        |
| Gated Conv FFN        | 147.50  | 19425.99   | 6832.47        | Competitive quality but much slower |

Table 7: Hybrid architecture results.

The convolution-augmented variants were evaluated using the strongest attention and positional configuration from the earlier experiments: GQA with ALiBi. The standard Transformer block remains the best overall trade-off, achieving the lowest validation perplexity while also using substantially less memory and providing higher throughput. Adding a convolution before attention hurts both quality and efficiency, increasing peak training memory from 4153 MB to 6453 MB and degrading validation perplexity from 146.24 to 173.83. The gated convolutional feed-forward block is much more competitive, reaching validation perplexity 147.50, but it is significantly slower and more memory-intensive than the standard block. Therefore, the hybrid experiments do not improve the final model choice, but they show that gated convolution is a plausible local-context augmentation whereas naive pre-attention convolution is not.

## 6 Analysis

The first major conclusion from the experiments is that not all efficient attention variants are equally useful in this setting. Sliding-window attention did not improve either quality or efficiency enough to justify its use at the tested budget, and the logged run should be interpreted as preliminary because it was trained for fewer steps than the other variants. The current implementation computes local key/value windows directly rather than materializing a full attention matrix, so its attention-score memory scales with sequence length times window size. Linear attention improved validation perplexity slightly over the standard baseline but required substantially more training memory and reduced throughput, which weakens its appeal as a practical replacement.

GQA, in contrast, improved both language modeling quality and efficiency. It reduced validation perplexity from 351.41 to 333.78 while slightly lowering peak training memory and maintaining very similar throughput. This makes GQA the most attractive attention choice among the tested variants.

The positional encoding experiments point even more strongly toward the importance of architecture choices for long-context behavior. ALiBi and RoPE both produce very large improvements when paired with GQA. ALiBi reduces validation perplexity from 333.78 for GQA + sinusoidal to a best logged value of 146.24, while RoPE reaches 150.28. Both curves show mild late-stage degradation after their best checkpoints, suggesting overfitting or diminishing returns rather than optimization instability. Relative positional bias reaches validation perplexity 235.86, which is a substantial improvement over sinusoidal but still well behind ALiBi and RoPE. In the current experiments, ALiBi gives the best positional encoding result, with RoPE as a close second.

Overall, the experiments suggest a useful hierarchy: GQA is the best attention mechanism tested so far, and ALiBi is the strongest positional encoding tested so far. This already gives a strong candidate model for the next stages of the project and for later hybrid convolution experiments.

The explicit extrapolation experiment strengthens this conclusion. A GQA + ALiBi model trained at context length 512 remains stable when evaluated at 1024 and 2048 tokens, reaching validation perplexity 147.76 at the longest tested context. In contrast, a matching GQA + RoPE model trained at 512 tokens degrades from 159.20 perplexity at 512 tokens to 280.24 at 2048 tokens, learned relative bias reaches 253.37 at 2048 tokens, and the standard sinusoidal baseline degrades much more severely from 300.04 to 1281.94. This separates in-context validation quality from length generalization: ALiBi is not merely the best positional encoding at the training length, but also the most convincing choice for extrapolating beyond it.

The convolution-hybrid experiments do not overturn this hierarchy. Gated convolution nearly matches the standard block’s validation perplexity, but its throughput and memory costs are much worse. The conv-before-attention design is clearly dominated in the current setup. This suggests that local convolutional mixing can be useful only when integrated carefully into the feed-forward path, and even then the marginal quality gain does not justify the compute overhead for this assignment-scale model.

## 7 Conclusion

The baseline standard Transformer with sinusoidal positional encoding establishes a reasonable starting point, but it is clearly not the strongest model in the current study. GQA emerges as the best attention variant, outperforming standard, sliding-window, and linear attention on validation perplexity while remaining efficient in both memory and throughput. ALiBi and RoPE further improve results substantially when combined with GQA, with ALiBi giving the best observed validation perplexity. The best logged GQA + ALiBi checkpoint occurs at step 7000, where the model reaches validation perplexity 146.24, while the final completed run at step 8500 remains close at 147.87. The best GQA + RoPE checkpoint reaches validation perplexity 150.28, making it a strong second-best positional encoding result. Convolution-hybrid variants do not improve the best model: the gated-convolution block is close in perplexity but much slower and more memory-intensive, while the conv-before-attention design performs worse.

The main lesson from the ablation study is that long-context modeling quality in this setup depends strongly on choosing the right attention and positional components rather than only scaling the baseline Transformer. Future work in the same codebase should add a fuller multi-context retraining matrix, especially at 4096 tokens if compute allows. The current results already support GQA + ALiBi as the final Core ML submission model.

## 8 Reproducibility Artifacts

The compact experiment artifacts used in this report are versioned under `results/core_ml/`. This includes final metric JSON files, available training-log tails, extrapolation outputs, latency benchmarks, and an aggregate `experiment_summary.json`. Large checkpoints and raw W&B/output directories are intentionally excluded from git to keep the repository small. The training script also supports mixed precision and gradient accumulation for future reruns, but the reported experiments use the settings stored in their logged configs and result artifacts.